Practical No.: 01

NAME: Bhalerao Tanvi Vitthal
ROLLNO.: 08
COURSE: AI&DS
CLASS: BE
SUB: Computer Laboratory-IV
Title: House Price Prediction Using Linear Regression.

1 House Price Prediction using Linear Regression

This notebook builds a simple Linear Regression model to predict house prices using a USA housing dataset. It is suitable for practical assessment use.

1.1 1. Import Required Libraries

[1]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics

# Display plots inside the notebook
%matplotlib inline
```

1.2 2. Load the Dataset

[2]:
```
data = pd.read_csv('USA_Housing.csv')
data.head()
```

[2]:
```
   Avg. Area Income  Avg. Area House Age  Avg. Area Number of Rooms  \
0      79545.458574             5.682861                   7.009188
1      79248.642455             6.002900                   6.730821
2      61287.067179             5.865890                   8.512727
3      63345.240046             7.188236                   5.586729
4      59982.197226             5.040555                   7.839388

   Avg. Area Number of Bedrooms  Area Population         Price  \
0                          4.09     23086.800503  1.059034e+06
1                          3.09     40173.072174  1.505891e+06
2                          5.13     36882.159400  1.058988e+06
3                          3.26     34310.242831  1.260617e+06
4                          4.23     26354.109472  6.309435e+05

                                             Address
```

```
0   208 Michael Ferry Apt. 674\nLaurabury, NE 3701…
1   188 Johnson Views Suite 079\nLake Kathleen, CA…
2   9127 Elizabeth Stravenue\nDanieltown, WI 06482…
3                                USS Barnett\nFPO AP 44820
4                               USNS Raymond\nFPO AE 09386
```

1.3 3. Dataset Information

[3] :

```python
data.info()

data.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 5000 entries, 0 to 4999
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype
---  ------                        --------------  -----
 0   Avg. Area Income              5000 non-null   float64
 1   Avg. Area House Age           5000 non-null   float64
 2   Avg. Area Number of Rooms     5000 non-null   float64
 3   Avg. Area Number of Bedrooms  5000 non-null   float64
 4   Area Population               5000 non-null   float64
 5   Price                         5000 non-null   float64
 6   Address                       5000 non-null   object
dtypes: float64(6), object(1)
memory usage: 273.6+ KB
```

[3] :

	Avg. Area Income	Avg. Area House Age	Avg. Area Number of Rooms	\
count	5000.000000	5000.000000	5000.000000	
mean	68583.108984	5.977222	6.987792	
std	10657.991214	0.991456	1.005833	
min	17796.631190	2.644304	3.236194	
25%	61480.562388	5.322283	6.299250	
50%	68804.286404	5.970429	7.002902	
75%	75783.338666	6.650808	7.665871	
max	107701.748378	9.519088	10.759588	

	Avg. Area Number of Bedrooms	Area Population	Price
count	5000.000000	5000.000000	5.000000e+03
mean	3.981330	36163.516039	1.232073e+06
std	1.234137	9925.650114	3.531176e+05
min	2.000000	172.610686	1.593866e+04
25%	3.140000	29403.928702	9.975771e+05
50%	4.050000	36199.406689	1.232669e+06
75%	4.490000	42861.290769	1.471210e+06
max	6.500000	69621.713378	2.469066e+06

1.4 4. Feature Selection

We remove non-numeric columns and separate input features (X) and target (y).

[4] :
```
X = data.drop(['Price', 'Address'], axis=1)
y = data['Price']
```

1.5 5. Train-Test Split

[5] :
```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.3, random_state=42
)

X_train.shape, X_test.shape
```

[5]: ((3500, 5), (1500, 5))

1.6 6. Train Linear Regression Model

[6]:
```
model = LinearRegression()
model.fit(X_train, y_train)
```

[6]: LinearRegression()

1.7 7. Model Prediction

[7]:
```
predictions = model.predict(X_test)
predictions[:5]
```

[7] :
```
array([1308536.13592605, 1237122.72746462, 1243835.62817079,
       1229241.70263994, 1059352.51821862])
```

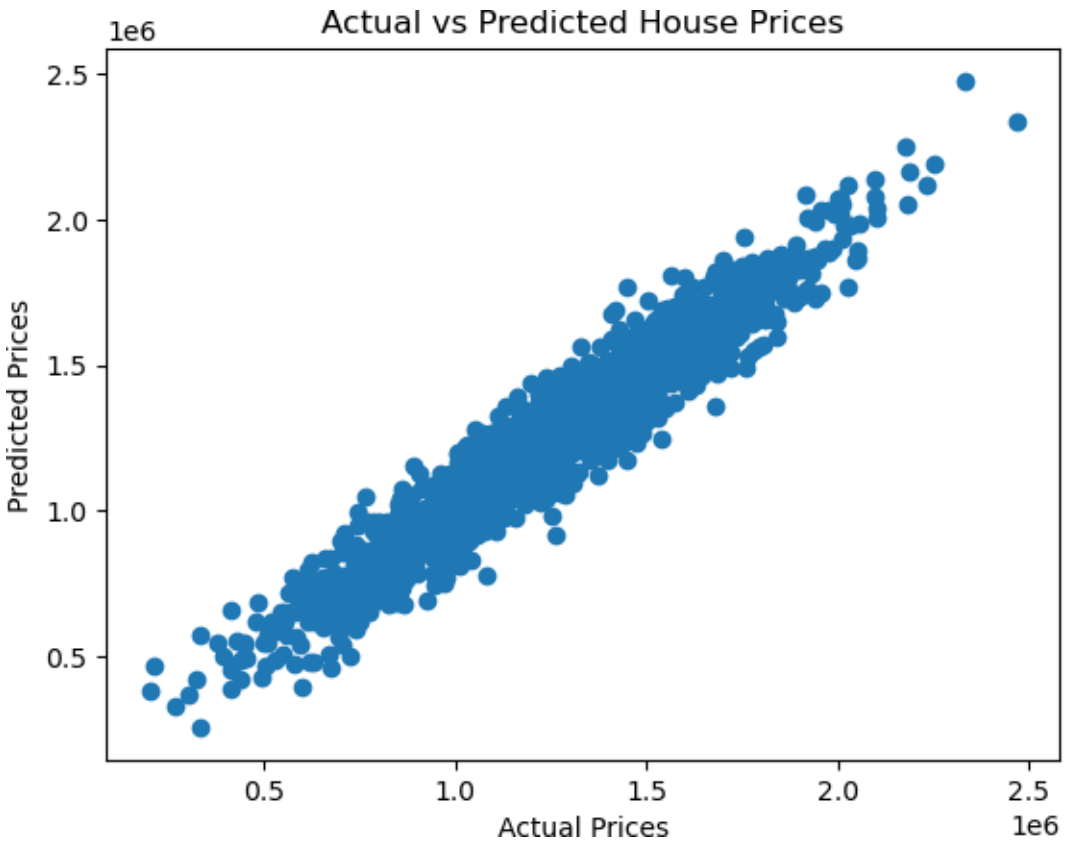
1.8 8. Model Evaluation

[8] :
```
print('MAE:', metrics.mean_absolute_error(y_test, predictions))
print('MSE:', metrics.mean_squared_error(y_test, predictions))
print('RMSE:', np.sqrt(metrics.mean_squared_error(y_test, predictions)))
```

```
MAE: 81135.5660933688
MSE: 10068422551.400925
RMSE: 100341.52954485458
```

1.9 9. Actual vs Predicted Prices

[9] :
```
plt.scatter(y_test, predictions)
plt.xlabel('Actual Prices')
plt.ylabel('Predicted Prices')
plt.title('Actual vs Predicted House Prices')
plt.show()
```


Actual vs Predicted House Prices
1e6
2.5
2.0
1.5
1.0
0.5
Predicted Prices
0.5
1.0
1.5
2.0
2.5
Actual Prices
1e6